

Human-Following Robot: Smart Assistance Using Ultrasonic and IR Sensors

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Abstract. Human-following robots are becoming increasingly important in service robotics because they can assist people by moving alongside them while avoiding obstacles and adapting to changing surroundings. These systems are finding applications in healthcare, logistics, industrial automation, retail environments, and personal assistance. Building a reliable robot, however, requires accurate human detection, continuous tracking, and safe navigation in dynamic environments.

This paper presents the design of a cost-effective human-following robot that primarily relies on ultrasonic and infrared (IR) sensors. The proposed system focuses on providing dependable human tracking using inexpensive hardware while maintaining a modular architecture that supports future upgrades such as cameras, LiDAR, or RGB-D sensors. Such flexibility makes the platform suitable for both educational purposes and real-world service applications.

In addition to describing the prototype, this work discusses several modern developments in human-following robotics, including sensor fusion, visual perception, trajectory prediction, simultaneous localization and mapping (SLAM), and socially aware navigation. Challenges such as target occlusion, obstacle avoidance, and long-term tracking are also examined along with possible solutions.

To evaluate the effectiveness of the proposed approach, both technical performance measures and user-oriented metrics are considered. The study demonstrates that a carefully designed low-cost sensing system can achieve reliable human-following behaviour while allowing straightforward integration of more advanced perception techniques when required.

Keywords: Human-Following Robot · Ultrasonic Sensors · Infrared Sensors · Sensor Fusion · SLAM · Socially Aware Navigation

1 Introduction

Human-following robots are autonomous mobile systems developed to detect, identify, and continuously follow a specific person while maintaining a safe and

comfortable distance. These robots have become an active area of research because they can simplify many routine tasks and improve efficiency in environments where humans and robots work together.

Today, human-following robots are used in hospitals for transporting medical supplies, in warehouses for assisting workers with material handling, in shopping centres as intelligent carts, and in personal assistance systems that support elderly or physically challenged individuals. Their ability to operate without constant human control makes them suitable for a wide range of practical applications.

Designing an efficient human-following robot is not a simple task. The robot must reliably recognize the intended user, estimate the person’s position in real time, and react appropriately to surrounding obstacles. Environmental conditions such as poor lighting, moving crowds, sensor noise, and temporary occlusions make this problem significantly more challenging.

Low-cost sensors such as ultrasonic and infrared devices provide an attractive solution for educational and budget-friendly robotic platforms. Ultrasonic sensors measure the distance between the robot and nearby objects, whereas infrared sensors help determine directional information for smoother movement. These sensors require little computational power, consume minimal energy, and can provide dependable performance in many indoor environments.

Although sensor-based systems perform well in structured environments, more demanding situations require additional sensing technologies. Cameras, RGB-D sensors, LiDAR, inertial measurement units, and simultaneous localization and mapping techniques can significantly improve tracking accuracy and navigation reliability. Therefore, a modular architecture that supports future hardware upgrades is highly desirable.

The primary objective of this work is to develop a practical and affordable human-following robot using ultrasonic and infrared sensors while maintaining sufficient flexibility for integrating advanced perception modules in the future. The proposed design emphasizes low cost, modularity, safe navigation, and ease of implementation without sacrificing reliability.

The remaining sections of this paper present the background and motivation behind the proposed system, describe the hardware and software architecture, discuss perception and navigation techniques, explain the evaluation methodology, and conclude with future research directions.

2 Background and Motivation

Early human-following robots mainly depended on simple techniques such as colour tracking, infrared beacons, or predefined markers to locate a user. Although these approaches were easy to implement, they worked reliably only in controlled environments with minimal distractions. Their performance decreased significantly when multiple people were present or when the target was temporarily hidden.

Recent developments in computer vision and embedded computing have greatly improved the capabilities of human-following robots. Modern systems combine cameras, depth sensors, LiDAR, inertial sensors, and deep learning algorithms to identify people more accurately and estimate their movement. Object detection models such as the YOLO family enable real-time person detection, while RGB-D cameras provide valuable depth information for safer navigation.

Despite these improvements, advanced sensing hardware considerably increases the overall system cost and computational requirements. Many educational institutions, small industries, and research laboratories require affordable robotic platforms that can perform reliably without expensive components. Therefore, low-cost sensing solutions based on ultrasonic and infrared sensors remain attractive for many practical applications.

Another important aspect of human-following robots is user acceptance. Even if a robot performs accurately from a technical perspective, users may hesitate to interact with it if its movements appear unsafe or unpredictable. Maintaining appropriate interpersonal distance, producing smooth motion, and behaving consistently all contribute to a better user experience. Consequently, human-centred navigation has become an essential consideration alongside traditional tracking accuracy.

In addition to usability, safety and privacy are equally important. A practical human-following robot should operate according to established robotic safety standards while minimizing unnecessary collection of personal information. Designing systems that balance affordability, safety, and performance remains one of the major goals of current research.

3 System Architecture and Methodology

The proposed human-following robot adopts a modular architecture that separates sensing, decision making, and motion control into independent functional layers. Such an organization simplifies maintenance, allows hardware upgrades, and enables future integration of more advanced perception modules without redesigning the complete system.

The architecture consists of three primary layers:

1. Sensing and Perception Layer
2. State Estimation and Mapping Layer
3. Motion Control and User Interaction Layer

Each layer performs a dedicated task while exchanging information with the remaining components through clearly defined interfaces. This modular design improves flexibility and overall system reliability.

3.1 Sensing Suite

The prototype primarily relies on inexpensive sensing devices capable of operating effectively in indoor environments.

- **Ultrasonic Sensors (HC-SR04)**

Multiple ultrasonic sensors are positioned around the robot to measure the distance between the robot and nearby objects. Their combined measurements help estimate the direction of the target while simultaneously detecting surrounding obstacles. These sensors offer reliable short-range measurements with very low computational overhead.

- **Infrared (IR) Sensors**

Infrared sensors complement ultrasonic sensing by providing fast directional information during movement. Their quick response allows the robot to perform small heading corrections and maintain smoother tracking performance.

- **Wheel Encoders and IMU**

Wheel encoders estimate the travelled distance, while the inertial measurement unit measures acceleration and rotational motion. Combining these measurements improves localisation accuracy and reduces the effect of noisy sensor readings.

- **Optional Vision Sensors**

Whenever higher accuracy is required, the platform can be extended using RGB cameras, RGB-D sensors, or LiDAR. These devices improve human identification, depth estimation, obstacle detection, and simultaneous localisation and mapping (SLAM).

3.2 Computing Platform

The proposed architecture distributes computational tasks across multiple processing units.

A microcontroller such as an Arduino performs real-time sensor acquisition and motor control. A Raspberry Pi or another single-board computer executes higher-level functions including behaviour planning, state estimation, and communication between software modules.

If computer vision or deep learning algorithms are integrated into the system, an embedded GPU platform such as NVIDIA Jetson can execute neural network inference while maintaining real-time performance.

3.3 Locomotion and Actuation

The robot uses a differential-drive mechanism powered by two DC geared motors. This configuration offers a simple yet reliable solution for indoor navigation while allowing smooth turning and accurate speed control.

Motor commands generated by the controller are transmitted through a motor driver such as the L298N module, which converts software commands into appropriate PWM signals for each motor. This arrangement enables precise control of both linear and rotational movement during human-following tasks.



Fig. 1. Prototype chassis and arrangement of ultrasonic and infrared sensors.

4 Perception: Detection, Tracking, and Re-Identification

Reliable perception is one of the most important requirements for a human-following robot. The robot must continuously identify the intended user, estimate the user’s position, and distinguish the target from surrounding people and obstacles. To achieve this, the proposed system combines low-cost sensors with optional vision-based techniques that can be integrated whenever higher accuracy is required.

4.1 Low-Cost Human Detection

The primary sensing mechanism consists of multiple ultrasonic sensors and infrared (IR) sensors mounted around the robot. Ultrasonic sensors continuously measure the distance between the robot and nearby objects, while IR sensors provide directional information that helps maintain alignment with the moving target.

Using measurements from several ultrasonic sensors positioned at the front and sides of the robot, the controller estimates the relative position of the person being followed. Distance variations indicate the movement of the target, allowing the robot to adjust both its speed and heading. This sensing strategy performs well in indoor environments where only a single person is present and the surrounding area is relatively structured.

Although inexpensive and computationally efficient, ultrasonic and IR sensors have certain limitations. Their performance decreases when several people

are located close together or when the target is temporarily blocked by obstacles. Therefore, additional sensing technologies may be incorporated for more demanding environments.

4.2 Vision-Based Detection

To improve perception accuracy, the system can optionally integrate computer vision techniques using RGB or RGB-D cameras.

Modern object detection algorithms such as the YOLO family provide accurate real-time human detection while maintaining high computational efficiency. When combined with depth information, detected objects can be converted into three-dimensional coordinates relative to the robot. This additional information significantly improves obstacle avoidance, path planning, and target localization.

Running these algorithms on embedded GPU platforms enables real-time inference while preserving the portability of the robotic platform.

4.3 Person Re-Identification

Long-term tracking requires more than simply detecting a person in every frame. When the target becomes temporarily hidden behind obstacles or moves through crowded environments, the robot must correctly recognize the same individual after reappearing.

Person re-identification techniques address this challenge by generating compact visual feature representations that describe a person’s appearance. These feature vectors remain relatively stable even when viewing angles or illumination conditions change.

The proposed system combines short-term motion tracking with appearance-based matching. Motion estimation predicts the expected position of the target, while appearance features verify the person’s identity after temporary occlusions. This combination reduces identity switching and improves overall tracking stability.

5 State Estimation and Occlusion Handling

Sensor measurements are naturally affected by noise and environmental uncertainty. Consequently, state estimation plays an essential role in maintaining smooth and reliable robot behaviour.

5.1 Kalman Filtering

Kalman filtering provides an efficient method for reducing measurement noise while estimating the position and velocity of the target.

Information obtained from ultrasonic sensors, inertial sensors, wheel encoders, and optional cameras is combined to generate a stable estimate of the user’s

movement. Even if sensor readings fluctuate slightly, the filter maintains smooth tracking by predicting the most probable target position.

For nonlinear robot motion, the Extended Kalman Filter (EKF) provides improved estimation accuracy by considering nonlinear system dynamics.

5.2 Learning-Based Motion Prediction

Temporary target loss frequently occurs in crowded environments where people continuously block each other’s movement.

To improve recovery performance, learning-based trajectory prediction models can estimate the future movement of the target using previous observations. Neural network architectures such as Long Short-Term Memory (LSTM) networks, Graph Neural Networks (GNNs), and Social-LSTM models are capable of predicting human trajectories by considering interactions between nearby individuals.

These predicted trajectories allow the navigation system to continue moving safely instead of stopping immediately whenever the target disappears for a short duration.

5.3 Occlusion Recovery Strategy

Whenever the robot temporarily loses the target, a multi-stage recovery strategy is activated.

1. Continue tracking using EKF prediction for a short period.
2. Navigate toward the last observed position of the target.
3. Perform a local search using rotational scanning.
4. Expand the search area using a spiral exploration pattern.
5. Notify the user or request manual intervention if the target cannot be located within the predefined timeout period.

This hierarchical recovery mechanism improves robustness while preventing unnecessary robot movement during prolonged target loss.

6 Navigation, Planning, and Social Compliance

After detecting and tracking the target, the robot must navigate safely while maintaining a comfortable distance from the user. The proposed navigation strategy combines motion control, obstacle avoidance, and socially acceptable behavior to ensure smooth operation in indoor environments.

6.1 Virtual Spring-Damper Controller

A virtual spring-damper controller is used to regulate the robot's movement relative to the target. The virtual spring generates an attractive force that pulls the robot toward the desired following position, while the damping component minimizes sudden acceleration and oscillatory motion.

This control strategy enables the robot to maintain a stable distance behind the user while adapting smoothly to changes in walking speed and direction. Controller parameters can be adjusted according to the robot's mechanical characteristics and the desired following distance.

6.2 Local Path Planning Using Dynamic Window Approach

The Dynamic Window Approach (DWA) is employed as the local path planner. Instead of selecting a path solely based on the shortest distance, DWA evaluates multiple feasible velocity combinations that satisfy the robot's motion constraints.

Each candidate trajectory is assessed using several criteria, including:

- Distance to the target
- Obstacle clearance
- Robot stability
- Smoothness of motion
- Direction toward the desired goal

The trajectory with the highest overall score is selected for execution. This approach allows the robot to move safely while continuously adapting to changes in the surrounding environment.

6.3 Socially Aware Navigation

When operating near people, navigation should consider human comfort in addition to collision avoidance. The proposed system therefore incorporates socially aware navigation principles.

The robot maintains an appropriate interpersonal distance, avoids cutting directly across walking paths, and minimizes sudden movements that may surprise nearby individuals. These behaviors improve user confidence and encourage natural interaction between humans and robots.

Additional penalties are included in the navigation cost function whenever the robot attempts to enter personal space or interfere with groups of people. As a result, the generated paths are smoother, safer, and socially acceptable.

6.4 Handling the Freezing Robot Problem

One common challenge in crowded environments is the freezing robot problem, where the planner cannot identify a collision-free path and the robot remains stationary.

Several strategies can reduce this issue:

- Predicting the future movement of nearby pedestrians before selecting a path.
- Allowing controlled deviations from the ideal trajectory while maintaining safety.
- Frequently replanning the path as new sensor information becomes available.
- Applying reinforcement learning methods that enable the robot to learn socially appropriate navigation behaviors from experience.

These techniques improve mobility and help the robot continue following the target even in congested environments.

7 Human–Robot Interaction

The success of a human-following robot depends not only on technical performance but also on user acceptance. A robot that behaves predictably and communicates its intentions clearly is more likely to be trusted by users.

Several design principles improve interaction quality.

7.1 Legible Motion

Robot movements should clearly indicate future actions. Gradual turns, smooth acceleration, and controlled deceleration allow users to anticipate the robot’s behavior, reducing confusion and improving safety.

7.2 Predictable Behaviour

Maintaining a consistent following distance and stable walking speed increases user confidence. Sudden speed changes or unnecessary directional adjustments should be avoided whenever possible.

7.3 System Feedback

Simple feedback mechanisms such as LEDs, indicator displays, or audio notifications inform users about the robot’s operating state. The system can indicate whether it is following, searching for the target, waiting, or stopped.

7.4 Personalization

Different users may prefer different following distances or walking positions. Allowing these preferences to be configured improves comfort and overall user satisfaction.

8 Evaluation Protocol and Performance Metrics

A comprehensive evaluation considers both technical accuracy and the quality of interaction with users.

8.1 Perception Performance

The perception module is evaluated using standard multi-object tracking metrics.

- **MOTA (Multiple Object Tracking Accuracy):** Measures the overall tracking performance by considering missed detections, false detections, and identity switches.
- **MOTP (Multiple Object Tracking Precision):** Measures the localization accuracy of successfully tracked targets.
- **Identity Switches:** Counts the number of times the tracking algorithm incorrectly changes the identity of the target.

8.2 System-Level Performance

Overall robot performance is measured using several practical metrics.

- Following distance error
- Robot reaction time
- Target recovery rate after temporary occlusion
- Navigation smoothness
- Number of obstacle collisions
- Success rate of completing the following task

8.3 Human-Centred Evaluation

Technical performance alone is insufficient for evaluating a service robot. User studies should therefore measure subjective factors including:

- Perceived safety
- User comfort
- Ease of interaction
- Confidence in robot behaviour
- Overall user satisfaction

These evaluations may be conducted using Likert-scale questionnaires after participants complete predefined navigation scenarios. Combining objective performance measures with user feedback provides a more comprehensive assessment of the proposed human-following robot.

9 Ethical, Safety, and Standards Considerations

The deployment of human-following robots should prioritize both user safety and responsible data handling. Since these robots operate in close proximity to people, their design must minimize the risk of accidents while maintaining reliable performance.

Whenever cameras or other vision sensors are used, only the information necessary for navigation and tracking should be processed. Sensitive personal data should not be stored unnecessarily, and all collected information should be protected using appropriate security measures.

The mechanical design should also include multiple safety mechanisms such as emergency stop buttons, obstacle detection sensors, reduced operating speeds in crowded environments, and collision avoidance algorithms. These features help prevent injuries and improve public confidence in autonomous robotic systems.

Whenever possible, the robot should follow internationally accepted collaborative robot safety standards such as ISO 10218 and ISO/TS 15066. Compliance with these standards improves reliability and facilitates deployment in industrial and service environments.

10 Case Studies and Deployment Scenarios

Human-following robots can be adapted for a wide variety of real-world applications.

10.1 Warehouse and Logistics

Warehouses often require workers to transport heavy materials over long distances. A human-following robot can automatically carry packages while following warehouse personnel, reducing physical effort and improving operational efficiency. Reliable obstacle avoidance and continuous tracking are particularly important in narrow warehouse aisles.

10.2 Healthcare

Hospitals and healthcare facilities can benefit from autonomous robots that accompany patients, transport medicines, or deliver medical equipment. In these environments, smooth navigation and predictable movement are essential for maintaining patient safety and comfort.

10.3 Retail and Shopping Assistance

Human-following robots can also function as intelligent shopping assistants. Autonomous shopping carts are capable of following customers throughout a store, allowing them to shop without pushing heavy carts manually. Similar systems may also assist elderly or physically challenged users in public spaces.

11 Open Research Directions

Although significant progress has been achieved in recent years, several research challenges remain open.

- **Continual Learning**
Future robots should continuously improve their perception models by learning from new environments without forgetting previously acquired knowledge.
- **Federated Learning**
Collaborative learning between multiple robots can improve overall performance while preserving user privacy because raw sensor data remains on individual devices.
- **Language-Based Interaction**
Large language models and multimodal AI systems may allow users to communicate naturally with robots using spoken language instead of predefined commands.
- **Formal Safety Verification**
Future navigation algorithms should provide mathematical guarantees regarding safe operation even in highly dynamic environments.
- **Group Following**
Instead of tracking a single individual, future systems should identify and follow groups of people while maintaining socially acceptable behavior.

12 Prototype Implementation Summary

The proposed prototype combines ultrasonic sensors, infrared sensors, an Arduino microcontroller, and a Raspberry Pi to implement a practical low-cost human-following robot.

Experimental observations indicate that the robot can successfully maintain a stable following distance under normal indoor conditions while responding quickly to changes in user movement. The integration of multiple sensors improves obstacle detection and reduces tracking errors compared with using a single sensing modality.

Although the baseline system primarily relies on inexpensive hardware, its modular architecture allows future integration of computer vision, LiDAR, SLAM, and deep learning algorithms without significant structural modifications. This flexibility makes the platform suitable for educational projects, research activities, and practical service applications.

13 Conclusion

This work presented the design and implementation of a low-cost human-following robot that combines ultrasonic and infrared sensors to achieve reliable target tracking and safe navigation. The proposed modular architecture separates sensing, state estimation, motion control, and user interaction into independent components, making the system easier to maintain and extend.

The study demonstrates that affordable sensing hardware can provide dependable human-following performance in structured indoor environments while maintaining low computational requirements. Additional perception modules

such as RGB cameras, LiDAR, or depth sensors can be integrated whenever higher tracking accuracy is required.

Beyond the hardware implementation, this paper also discussed several important topics related to modern human-following robots, including sensor fusion, trajectory prediction, socially aware navigation, human-robot interaction, and safety considerations.

Future work will focus on improving long-term tracking, integrating advanced artificial intelligence techniques, enhancing human-robot communication, and developing more robust navigation strategies for crowded environments.

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